Asvinth K

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.33
PAAVAI VIDHYASHRAM	Class XII (CBSE)	2023	90.80%
GREEN PARK	Class X (CBSE)	2021	89.20%

EXPERIENCE

Software | Agentic AI Intern | VH Mobile Metrics

- **Engineered an end-to-end subscription intelligence platform** using **ASP.NET Core, Python, LangGraph, PostgreSQL/pgvector, and Flutter**, enabling automated detection and tracking of recurring payments from **Account Aggregator (AA) bank transaction data**.
- **Developed a hybrid deterministic + LLM reasoning pipeline** using **merchant normalization, RapidFuzz, TF-IDF, BGE embeddings, clustering, confidence scoring, and weighted LLM fusion** to identify subscriptions, maintain lifecycle states, and provide explainable detection evidence.
- **Implemented the production application stack and intelligent workflows** with **Docker, Node.js/Wrangler, Cloudflare Workers/Containers, GitHub Actions, JWT authentication, analytics, notifications, budgeting, and an LLM-powered cancellation agent**, with automated validation through **xUnit and pytest**.

PROJECTS

Research Project - Multimodal LLMs for 3D Brain MRI Segmentation and Disease Diagnosis

- **Adapted and deployed** BiomedParse for **3D brain MRI segmentation**, reducing false positives via **multi-plane probability fusion**, and extended it to **IIT-BHU's 3D MRI datasets** for improved anatomical coverage.

Restormer-Based Image Restoration Pipeline | *KLA AI Hack × IIT-H 2026*

- **Built** a 2× super-resolution + denoising pipeline with a pretrained **Restormer backbone** and custom **RRDB + channel-attention head**, using staged freezing/unfreezing for fine-tuning.
- **Scaled** training across dual Kaggle T4 GPUs with **Accelerate/DDP**, progressive patch curriculum, EMA/SWA, and TTA, improving PSNR/SSIM from **22.9 dB / 0.56** to **28.5 dB / 0.79**.

Automated ID-Card Data Digitization Pipeline (Gemini-Powered OCR SaaS)

- **Identified** a manual ID-card data-entry bottleneck (~50,000 forms/year) and **built** an automated AI pipeline to replace manual re-typing.
- **Achieved 96% extraction accuracy at 45 forms/minute** using an async **FastAPI job queue**, batched Gemini API calls, and rate-limited round-robin key pooling with direct Excel output.
- **Designed** a multi-tenant SaaS with session authentication, business-hour gating, and batch isolation, priced at **1/form (~2× cheaper, ~200× faster)** than manual typing.

Interpretable AI Framework for ECG Sheet Digitization & Cardiac Diagnosis | *Inventor, Patent Applied* -- IITH

- **Fine-tuned** a **YOLOv8n** detector to crop all 12 ECG leads, achieving **99.8% precision** and **100% recall** across lead classes.
- **Built** two independent digitization models (**CNN**- and **Mamba**-based) using a novel **position-weighted softmax localization** technique for 2D->1D ECG signal conversion, each reaching accuracy **on par with global benchmarks**.
- **Combined** both into a **Hybrid-MC architecture**, negating each model's individual weaknesses to reach **97.8% correlation / 37.1 dB PSNR**, with markedly better robustness on **zoom-augmented data** where standalone models degraded.
- **Trained and deployed the full pipeline end-to-end on Kaggle**, achieving strong results using just **4,000 of 22,000 available samples** due to compute constraints.
- **Developed** a transformer-CNN-Mamba **ECG diagnoser** classifying 5 cardiac superclasses, reaching **0.90 macro / 0.92 micro ROC-AUC**, comparable to pre-published benchmark models.

Mini-ML Library & Autograd Engine | *Course: FOML -- Prof. Konda Reddy Mopuri*

- **Built a Mini-ML library** with **scikit-style APIs** for models, preprocessing, dataset loading, and model-selection utilities.
- **Implemented a NumPy-based Autograd Engine** with **computation graphs, automatic differentiation**, matrix operations, activations, and reductions.
- **Designed a modular neural-network framework** with Linear, ReLU/Sigmoid, Sequential, SGD, and Cross-Entropy components powered by the custom autograd backend.
- **Engineered ML pipelines** on **Spambase and Fashion-MNIST**, including **IRLS Logistic Regression**, feature engineering, hyperparameter tuning, and reproducible benchmarking.

SKILLS

Languages: Python, C++, Java, SQL, Bash

ML/AI: PyTorch, Transformers, Hugging Face, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas

Tools: Git, Docker, PostgreSQL

RELEVANT COURSES

Core Math Courses: Calculus-I & II, Intro to Statistics, Intro to Probability, Linear Algebra, Linear Optimization, Game Theory

Applied Mathematics Courses: Probability and Random Variables, Transform Techniques, Differential Equations, Numerical Methods, Complex Variables, Applied Math for chemical engineering, Discrete Mathematics

Computer Science Courses: Intro to Programming, Data Structures and Algorithms, Algorithms, Operating Systems, Software Development Fundamentals, Introduction to Computing, **DBMS,** Data Mining, Distributed Ai Training on Large-scale GPU Clusters

ML courses: FOML, Concentration Inequalities, Computer vision, AI for Drug Discovery, NLP, MedicalAI -Prototyping and Development, Machine Learning Specialization by Andrew Ng, issued by Stanford University.

EXTRACURRICULAR

- Won **3 medals** in Milan and Diesta inter-college powerlifting competitions; active **NSS volunteer** and participant in a **Musi River cleanup** organized by the Suzuki Innovation Centre

POSITIONS OF RESPONSIBILITY

Finance and equipment team core in EBSB cultural appreciation group of IITH